

SIMATS ENGINEERING
ASSESSMENT TOOL 3: Reverse Engineering Task

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1507

COURSE OUTCOME COVERED

CO4: *Analyze big data characteristics and challenges across domains including security and application aspects.*
(BL4)

Dave's Taxonomy: Articulation (Level 4)
Question Count: 5

Total Marks: 30

Code of Conduct

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: _____

Assessment Tasks

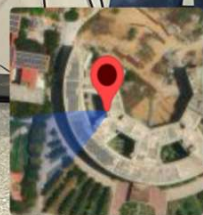
Reverse Engineering Task

For each task, study the described big data solution, infer how it was built, identify the underlying components, and explain what design decisions were likely made.

1. A big data platform collects billions of website clicks, mobile interactions, and customer transactions every day. Reverse engineer the probable distributed storage system, ingestion framework, and batch-processing tools used. Infer how structured and semi-structured data are separated and indexed for analysis. Identify the likely stream-processing engines supporting real-time analytics and reporting. Explain the scalability and partitioning strategies probably implemented for handling massive workloads. Describe how dashboards and business intelligence systems are likely connected to the analytics layer.
2. An e-commerce platform tracks cart additions, abandoned checkouts, product views, and payment behavior across regions. Reverse engineer the likely event streaming tools, customer profiling databases, and recommendation workflow used. Infer how session data, clickstream logs, and transaction records are synchronized for real-time personalization. Identify probable caching systems, distributed query engines, and machine learning stages involved. Explain how structured order data and semi-structured browsing data are merged for analytics. Deduce the scalability and fault-tolerance decisions likely implemented. Describe how dashboards and KPI monitoring are probably generated from the pipeline.
3. A big data healthcare system processes patient histories, diagnostic images, wearable sensor streams, and prescription records. Reverse engineer the probable hybrid database architecture managing structured and unstructured healthcare information. Infer the likely data integration tools connecting hospitals, laboratories, and insurance systems. Identify the streaming and machine learning frameworks probably used for disease prediction and monitoring. Explain how privacy,

encryption, and compliance requirements are likely enforced in the platform. Describe the analytics pipeline that may support clinical decision-making and population health studies.

4. A smart city platform gathers traffic sensor feeds, CCTV metadata, weather updates, and public transport activity continuously. Reverse engineer the probable edge computing setup, distributed messaging system, and centralized storage design. Infer how geospatial analytics and streaming computations are handled for congestion prediction. Identify the likely databases used for time-series and location-aware queries. Explain how batch and real-time analytics are combined for urban planning insights. Deduce the security and access-control measures likely protecting citizen and infrastructure data. Describe how visualization systems probably deliver live operational intelligence.
5. A healthcare analytics system integrates patient records, wearable device readings, lab reports, and appointment histories. Reverse engineer the probable hybrid storage architecture handling sensitive structured and unstructured medical data. Infer the likely ETL pipelines and interoperability standards connecting hospitals and clinics. Identify how real-time monitoring and predictive health analytics are probably implemented. Explain the role of distributed processing frameworks in population-level analysis. Deduce the encryption, compliance, and audit mechanisms likely enforced for data governance. Describe how machine learning models may support diagnosis risk scoring and treatment recommendations.



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GPS Map Camera



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E-Commerce Real-Time personalization pipeline.

It includes cart additions, abandoned checkouts, product views and payment behavior would typically use a real-time data pipeline combining streaming, caching

(i) Event streaming Tools:

- * product views
- * Add-to-cart events
- * checkout events
- * payment events
- * User clicks

(ii) Customer profiling Database

customer profiles may be stored in databases such as Redis, MongoDB, Cassandra or DynamoDB.

Real-Time Personalization

"Real-time personalization means giving recommendations based on what the customer is doing at that moment. For example, if a customer views or adds a shoe to the cart, the system immediately collects this activity. Then, the machine learning model analyzes it and recommends similar products. So, it helps customers find useful products quickly and also increases sales."

Synchronization of Data

"Synchronization means connecting different customer data together. For example, we can connect the customer's product views, cart activity, checkout, and payment details using customer ID or session ID. This helps the company understand the complete customer journey from browsing to purchasing. It also helps in providing better personalized services."

- Conversion rate
- Cart abandonment rate
- Average order value
- Product views
- Revenue
- Payment success rate
- Customer retention
- Recommendation click-through rate
- Regional sales
- Active users.

Scalability and Fault Tolerance

The platform would likely implement:-

- * Horizontal scaling by adding more servers.
- * Kafka partitioning for parallel event processing.
- * Replication of important data.
- * Load balancing across application servers.
- * Database replication.
- * Automatic service recovery.
- * Checkpointing in stream-processing systems.
- * Retry mechanisms for failed events.
- * Multiple availability zones or regions.

These mechanisms allow the platform to continue operating even if individual servers or services fail.

10. Dashboards and KPI Monitoring:-

Processed data can be sent to visualization tools such as power BI, Tableau, Grafana or Looker.

Important KPIs may include.

KPI should be
specific, time bound, Relevant,
measurable, Attainable

6) Machine Learning
Machine Learning is used in e-commerce platform to understand customer behavior and recommended suitable products.

for example

If a customer frequently repairs laptops and add laptops superior to the cost, the ML system learns this behavior and recommended similar products.

Workflow

Data collection → Data cleaning

→ Feature Engineering - Model

Training → Model Validation

Recommendation generation

1) Data collection

First the system collects customer activity from the e-commerce platform.

2) Data cleaning

The collected data may contain duplicate, incomplete or incorrect information therefore missing values and corrects errors.

3) Feature engineering
Feature engineering means converting raw customer activity into useful information called features

4) Model training
The cleaned data and useful features are given to a machine learning model

5) Model validation
checks whether the model gives accurate predictions

6) Recommendation
Suggests suitable products to the customers

7) Real time Prediction
Gives recommendation based on the customer's current activity

8. Feedback
customer clicks, cart additions and purchases are used to improve the model

example

If the customer views laptop repeatedly the system may recommend a laptop bag or wireless mouse